



No math this time - just clues! Each clue removes possibilities until only one match can be left.

MISSION: The Pet Clue Mystery

The problem

Alex, Ben, and Chloe each have a different pet: a dog, a cat, and a fish. Chloe has the fish. Alex does not have the dog. Ben does not have the cat. Who has each pet?

CLUE 1 - What is the problem asking you to find?

Say it in your own words. No calculating yet.

CLUE 2 - What do you KNOW for sure?

Start with the clue that gives an exact match. Then notice what that fact makes impossible for the other people.

YOUR FIRST PATH - What will you try?

Make a tiny table, cross out impossible pets, draw the three people, or reason in words. Choose your detective path.

THINKING TOKENS



I used a sure fact



I crossed out clues



I found what was left



Detectives do not guess when clues can do the work. Use one clue, see what changes, then use the next.

Use hints one at a time.

Tip for printing: cover each hint with a sticky note or a strip of scrap paper. Reveal only the next one you need.

HINT 1 - START WITH A SURE MATCH

One clue tells you exactly which pet belongs to one person. Which clue is that?

HINT 2 - REMOVE WHAT IS USED

If Chloe has the fish, can Alex or Ben also have the fish? What pets are still possible for them?

HINT 3 - USE A NOT CLUE

Alex does not have the dog. Of the pets still available to Alex, which one must be his?

HINT 4 - WHAT IS LEFT?

After you know Chloe's pet and Alex's pet, only one pet remains for Ben.

BACK TO YOUR PATH

Now try again. What did the hint help you notice?



Clue trail complete! Show how one fact led to the next before writing the final matches.

1. Show your path

Show the clues you used and what each clue ruled out. Cross-outs, a table, or sentences all work.

2. Your answer

Who has the dog, the cat, and the fish? _____

3. Another door?

Can you solve it again with a different representation - for example, a 3-by-3 grid instead of sentences?

My second idea:

4. Choose the Thinking Moves you used

- ☐ Started with a sure fact ☐ Ruled things out
- ☐ Used what was left ☐ Checked every clue

Purpose of this mission

This mission practices logical elimination without relying on arithmetic. The goal is for Amira to begin with a certain fact, use it to remove impossible choices, and keep chaining facts until only one consistent solution remains.

What to praise

- "You started from something you knew for sure."
- "You used a clue to rule out possibilities instead of guessing."
- "You noticed that one answer changed what was possible elsewhere."
- "You checked the final matches against every clue."

Answer

Chloe has the fish, Alex has the cat, and Ben has the dog.

Valid reasoning paths to recognize

Chain the clues

Chloe = fish. That leaves dog and cat for Alex and Ben. Alex is not dog, so Alex = cat. Ben must be dog.

Use a possibility grid

Make rows for Alex, Ben, Chloe and columns for dog, cat, fish. Mark Chloe-fish as yes, cross out fish for Alex and Ben, then cross out dog for Alex. The remaining matches are Alex-cat and Ben-dog.

Use what is left

Lock in Chloe-fish first. For Alex, fish is unavailable and dog is forbidden, so cat is the only choice. Dog is then the only pet left for Ben.

If she gets stuck

Ask: "Which clue gives us something we know for sure?" Once Chloe-fish is placed, ask: "What becomes impossible now?" Keep the conversation about eliminating possibilities rather than supplying the next match.